Agenda

1. Announcements

   - 11.4, 11.5, 11.6, 11.7 HW due Wednesday
   - Discussion Quiz 8 on Thursday over
   - Midterm 2 is next Tuesday April 11 at 7:30 pm

     Exam Info Page: go.wisc.edu/1xxzfx

2. Section 11.8 - Power Series

Power Series

An infinite series we make as a function.

Basically, an infinite polynomial.

Definition:

An expression for a power series centered at $X=0$ is

$$\underbrace{\sum_{n=0}^{\infty} a_n X^n}_{\text{coefficient of } n^{th} \text{ power of } X} = a_0 + a_1 X + a_2 X^2 + a_3 X^3 + \dots$$

where X is it's variable, and a_n's are it's coefficients. like a polynomial.

Activity: Write out the first four terms of the power series

1.) $\sum_{n=0}^{\infty} \frac{X^n}{n!} = \underset{n=0}{1} + \underset{n=1}{X} + \underset{n=2}{\frac{X^2}{2}} + \underset{n=3}{\frac{X^3}{6}} + \dots$

2.) $\sum_{n=0}^{\infty} \frac{X^n}{2^n} = \frac{X^0}{2^0} + \frac{X^1}{2^1} + \frac{X^2}{2^2} + \frac{X^3}{2^3} + \dots$

$= \underset{n=0}{1} + \underset{n=1}{\frac{X}{2}} + \underset{n=2}{\frac{X^2}{4}} + \underset{n=3}{\frac{X^3}{8}} + \dots$

Example: Write the power series in the form $\sum_{n=0}^{\infty} a_n x^n$

$1 - \frac{X}{3} + \frac{X^2}{5} - \frac{X^3}{7} + \dots$

$1 \cdot X^0 - \frac{1}{3} \cdot X^1 + \frac{1}{5} X^2 - \frac{1}{7} \cdot X^3$

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2n+1}$$

notice:
- changing sign

Sometimes it is advantageous to center a power series somewhere other than $x=0$:

<u>Definition</u>: An expression for a power series centered at <u>$x=c$</u> is $\sum_{n=0}^{\infty} a_n(x-c)^n$.

cannot take $5x$, would need to refactor.

<u>Activity</u>: Find the center of each power series.

1.) $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$ $c=3$

2.) $\sum_{n=0}^{\infty} \frac{(x+2)^n}{\sqrt{n+1}}$ $c=-2$

3.) $\sum_{n=0}^{\infty} \frac{(-1)^n}{2^n} x^n$ $c=0$

4.) $\sum_{n=1}^{\infty} \frac{(3x-2)^n}{n3^n}$ $c=2/3$

$3^n\left(x-\frac{2}{3}\right)^n$ factor

$3x-2=0$ or set $x=0$

$x=2/3$

Radius and Interval of Convergence

Example: For what x-values does $\sum_{n=0}^{\infty} x^n$ converge?

– $\sum x^n$ is a geometric series with $r = x$

- converges when $|r| = |x| < \boxed{1}$
- diverges when $|r| = |x| \geq 1$

radius of convergence

"how far away can we get to still converge?"

converges for all x in $\boxed{(-1, 1)}$.

interval of convergence

Example: For what x-values does $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converge?

ratio test!

Ratio Test!

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \lim_{n\to\infty} \left| \frac{x^{n+1}}{x^n} \cdot \frac{n!}{(n+1)!} \right|$$

$$= \lim_{n\to\infty} |x| \cdot \cancel{\frac{1}{n+1}}^{\,0} = 0 < 1$$

$\therefore$ converges

interval of convergence: $(-\infty, \infty)$

radius of convergence: ∞

Activity For what x-values does $\sum_{n=0}^{\infty} n!(x-2)^n$ converge?

$$\lim_{n\to\infty} \left| \frac{(n+1)!(x-2)^{n+1}}{n!(x-2)^n} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(n+1)!}{n!} \cdot \frac{(x-2)^{n+1}}{(x-2)^n} \right|$$

$$= \lim_{n\to\infty} (n+1) \cdot |x-2| = \infty, \; n \neq 2 \; > 1$$

The series diverges with $x \neq 2$

$\rightarrow \sum n!(x-2)^n$ only converges at $x = 2$.

IoC: $\{2\}$
R.C: 0

1) Converge everywhere
2) converge at a single point
3) converge at some subset

<u>Theorem</u>: Suppose $\sum a_n(x-c)^n$. There are three possible cases:

1.) The series converges <u>only at $x=c$</u>. $R=0$ $IoC = \{c\}$

2.) The series converges <u>everywhere</u> $R=\infty$ $IOC = (-\infty, \infty)$

3.) For some value $R>0$, the series converges for all x where $|x-c|<R$ and diverges for all x where $|x-c|>R$.

R is the radius of convergence

$-R < x-R < R$

$c-R < x < c+R$

IOC possibilities:

① $(c-R, c+R)$

② $[c-R, c+R)$

③ $(c-R, c+R]$

④ $[c-R, c+R]$

<u>Example</u>: Find the radius and interval of convergence of $\sum_{n=0}^{\infty} \frac{(x-3)^n}{n^2}$